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Enhancing Groundwater Level Forecasting using Hybrid Machine Learning Techniques

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***Abstract*—The rapid urban growth, intensive agricultural practices, and changes in climate are putting pressure on groundwater resources. Effective water resource planning and management therefore requires reliable prediction of the groundwater levels. Nevertheless, the correct prediction is not easy due to the nonlinear behavior of groundwater time-series, seasonal variations, and noise that depend on the effect of hydro-climatic and anthropogenic factors. In order to overcome these issues, this paper will propose a hybrid groundwater forecasting model, which combines Principal Component Analysis (PCA), Discrete Wavelet Transform (DWT), and Long Short-Term Memory (LSTM) networks. PCA is used to cut dimensions and eliminate redundancy between hydro-climatic variables, whereas DWT is used to denoise groundwater signals and respond to multi-scale temporal variations. The extracted features are then employed to train an LSTM model that can learn long-term temporal relationships on the dynamics of groundwater. The suggested framework is tested with the help of long-term multi-station groundwater observations. The standard regression measures such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R2) are used to measure model performance. The experimental findings reveal that the hybrid PCA-DWT-LSTM model has a better forecasting performance and robustness than the traditional standalone models, which means that it may be useful in groundwater monitoring and decision-support processes.**

***Index Terms*—Groundwater Level Forecasting, Hybrid Machine Learning, Principal Component Analysis, Discrete Wavelet Transform, Long Short-Term Memory, Time Series Prediction**

I. Introduction

Groundwater is an important source of freshwater which is used in irrigation of agriculture, domestic and industrial operations in most parts of the world. Over the past few decades, the high rate of population increase, extensive farming activities, and the growing climate variability have put a lot of pressure on the groundwater resources, which has resulted in the constant degradation of the aquifer systems. The capacity to predict the groundwater level under different hydro-climatic and man-made circumstances is thus critical in sustainable groundwater management.

The level of groundwater is a hard task to predict because of the complicated nature of the subsurface hydrological processes. Groundwater systems are highly nonlinear, have long-term temporal interactions, and seasonal fluctuations. Moreover, several correlated factors affecting the dynamics of groundwater include rainfall, temperature, evapotranspiration, recharge, and pumping activities. The conventional statistical models such as linear regression and autoregressive models are based on linear assumptions and stationarity, which restricts their applicability to the real world ground water data. Physical based hydrology models are interpretable, but demand a lot of domain knowledge, parameterization, and computational resources, and are not suitable to large scale or real time forecasting.

The latest developments in machine learning have brought about data-driven methods that can be used to model nonlinear relationships. Nevertheless, individual machine learning models are usually sensitive to noise, high-dimensional input space, and non-stationary signals that are typically observed in groundwater time series. In order to address such constraints, this paper suggests a hybrid groundwater level prediction model that combines Principal Component Analysis (PCA), Discrete Wavelet Transform (DWT), and Long Short-Term Memory (LSTM) networks. The proposed framework will enhance the accuracy and robustness of prediction by dimensionality reduction, signal denoising, and temporal sequence learning, which will be a reliable decision-support tool in sustainable groundwater management.

This work is novel in that it combines the use of PCA-based dimensionality reduction and DWT-based signal decomposition before LSTM temporal modeling. This mixed design is a solution to feature multicollinearity, noise reduction, and long-term temporal dependency learning in groundwater time-series prediction.

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II. Literature Survey

Deep learning, particularly Long Short-Term Memory (LSTM), is rapidly becoming a popular choice in recent groundwater level (GWL) forecasting research due to its ability to model nonlinear dynamics and temporal dependencies compared to most classical statistical models. Thakur and Sharma [1] showed that LSTM is able to detect seasonal and long term groundwater trends directly based on historical time-series. Similar findings were reported by Liu et al. [12], who demonstrated that deep recurrent architectures can effectively capture temporal dependencies in groundwater systems influenced by climatic variability. Nevertheless, time-only LSTM models can perform poorly in the presence of intense noise, sharp regime changes, or multi-scale periodicity typical of hydro-climatic data. To solve this, hybrid decomposition-deep learning approaches have emerged as a leading trend. Nazari et al. [2] combined empirical mode decomposition (EMD) and wavelet-based LSTM to isolate intrinsic oscillations and complexity reduction prior to prediction, which they claimed to be more stable and accurate, particularly in mixed-frequency groundwater variations. Liang et al. [4] similarly used wavelet transform and LSTM to improve learning by revealing frequency-localized features. Recent work by Sun et al. [13] further confirmed that wavelet-based hybrid models can significantly improve groundwater forecasting performance by separating noise from useful hydrological signals. Wu et al. [6] suggested a wavelet-LSTM model to model groundwater and favored the assertion that multi-resolution decomposition aids LSTM to learn both brief fluctuation and long-term patterns; these techniques are particularly effective in case the data under examination are periodic, absent, or sudden change due to pumping or rainfall aberration.

Another theme that is recurrent in multivariate groundwater forecasting is dimensionality reduction, particularly where groundwater is sensitive to a variety of climatic and anthropogenic drivers. A PCA-based multivariate LSTM of groundwater records under human influence was proposed by Kim et al. [3], which demonstrated that PCA can condense correlated inputs and reduce redundancy prior to sequence learning, and which works well when the effective degrees of freedom are few, yet the number of explanatory variables is large. Similarly, Zhang et al. [7] investigated the issue of groundwater level calculation with PCA and LSTM and confirmed the hypothesis that feature compaction with PCA can enhance the efficiency and robustness of learning by eliminating collinearity and emphasizing the most important modes. In addition to accuracy, uncertainty awareness and climate teleconnections are becoming a more popular topic of discussion as key to reliable decision-making. Chu et al. [5] simulated daily groundwater levels with LSTM and included teleconnection pattern data via PMI and used bootstrap-based methods to measure uncertainty, emphasizing that deterministic point forecasts might not be adequate to support risk-sensitive water management.

Although the literature on groundwater is mostly groundwater-centric, i.e. LSTM and decomposition hybrids, neighboring time-series areas offer supportive evidence to ensemble learning and hybrid statistical-ML designs in the presence of noisy environmental variability. More broadly, deep learning approaches have been widely applied in hydrological forecasting problems such as streamflow and groundwater prediction, demonstrating strong performance in capturing nonlinear environmental dynamics [14]. Nagesh et al. [8] investigated time-series prediction of rainfall with a mixture of SVM, KNN, ARIMA and E-GARCH, which is a larger trend: complementary learners can often be more robust to heteroscedasticity and abrupt changes in variance - qualities that groundwater and rainfall signals often exhibit. In terms of feature engineering and variable-selection, Chowdary and Radhika [9] highlighted the best variable identification and statistical mining, which supports the usefulness of the practicality of selecting compact and informative representations prior to training predictive models, which is similar to PCA-based reduction in hydro-environmental modeling. Equally, Reddy et al. [10] demonstrated that machine-learning pipelines to identify anomalies are sensitive to preprocessing, feature treatment, and model selection, which is also true of groundwater forecasting where sensor noise, outliers, and regime changes may confound learning systems unless addressed in a systematic manner.

In general, existing literature suggests three major observations: (i) LSTM can be successfully applied to predict groundwater time-series, but is susceptible to noise and multi scale structure when applied in isolation [1], [12]; (ii) decomposition algorithms such as wavelets or EMD enhance predictive learning by decomposing signals into more learnable components [2], [4], [6], [13]; and (iii) PCAbased feature compaction improves multivariate groundwater modeling when hydro climatic drivers are correlated [3], [7]. These findings are also supported by recent review studies on machine learning applications in groundwater forecasting [15].

III. Proposed Framework

The hybrid PCA-DWT-LSTM framework to groundwater level (GWL) forecasting is proposed in this paper, which is aimed at overcoming the inherent issues of hydrogeological time-series data: nonlinearity, multicollinearity, noise, and multi-scale temporal variations. Instead of using a single predictive model, the suggested framework splits the forecasting task into a series of structured steps, with every part of the proposed framework having a specialized role to enhance robustness and accuracy of prediction.

First, raw groundwater and hydro-climatic variables are processed and normalized to guarantee numerical stability and similar scales of features. Given that the groundwater levels depend on various correlated environmental and anthropogenic variables, the first transformation step is Principal Component Analysis (PCA). PCA is an orthogonal representation of the original high-dimensional input space in a smaller set of orthogonal principal components which represent much of the data variance. This action is a successful way of eliminating multicollinearity, minimizing redundancy between explanatory

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variables, and simplifying the learning space of the downstream model.

After dimensionality reduction, the signal of the groundwater level is subjected to the Discrete Wavelet Transform (DWT) to deal with non-stationarity and noise. Time-series of groundwater usually have seasonal cycles, spikes and spikes, and measurement noise. DWT breaks the signal into approximation and detail coefficients at a series of resolution levels, which allows to effectively suppress noise and preserve long-term trends and seasonal data, which are vital in making accurate forecasts.

The feature representation is then compact and denoised and fed into a Long Short-Term Memory (LSTM) network. The rationale behind the choice of LSTM is that it has a gated memory structure which allows it to model long-range temporal dependencies and sequential patterns, which are challenging to model with traditional recurrent or statistical models. The LSTM concentrates on learning significant temporal dynamics as opposed to training noise or redundant correlations by operating on PCA-reduced and wavelet-denoised inputs.

The Fig. 1 shows the architecture of the proposed groundwater level forecasting framework. The framework combines data pre processing, dimensionality reduction, signal processing, and deep learning algorithms to enhance the accuracy of prediction.

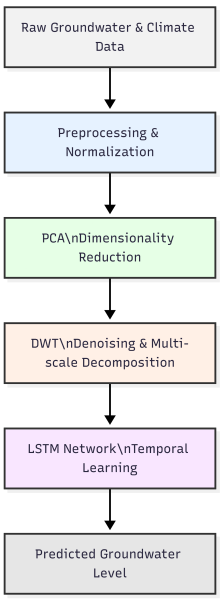


Fig. 1. Architecture of the proposed PCA–DWT–LSTM groundwater level forecasting framework.

The suggested frame-work will be a pipeline model that will enhance the accuracy of ground water level forecasting. To begin with, monitoring stations are used to measure raw groundwater and other related hydro climatic variables like rainfall, temperature, and humidity. These are the data that make up the input layer of the forecasting system.

The second stage involves preprocessing and normalization of the input features to eliminate inconsistencies and scale the variables equally. This measure enhances the stability of the model and easy training of the learning model.

Principal Component Analysis (PCA) is then used to do dimensionality reduction. PCA eliminates the redundant and highly correlated hydro-climatic variables and conserves the most informative elements of the data. This transformation makes the models less complex and increases the learning efficiency.

Once the dimensionality reduction has been applied, the Discrete Wavelet Transform (DWT) is applied to the groundwater time series in order to denoise it and break the signal down into several frequency components. The process aids in capturing multi-scale time pattern that exists in hydrological systems and rejects noise and irregular variations.

These transformed features are then fed into a Long Short-Term Memory (LSTM) network, which is trained to learn long-term temporal correlations and seasonal changes in the levels of groundwater. The LSTM is especially useful in modeling sequential hydrological data because it is capable of storing historical information over time.

Lastly, the trained network produces the predicted values of the groundwater level. The proposed PCA-DWT-LSTM architecture is a powerful framework to predict the groundwater level by integrating dimensionality reduction, signal decomposition, and deep temporal learning.

IV. System Implementation and Methodology

This part outlines the data preparation, feature transformation, model structure and training process used in the suggested hybrid PCA-DWT-LSTM groundwater forecasting framework.

A. Data Preparation and Preprocessing

The data in this paper is based on the quarterly ground water level readings of the monitoring stations that are run by the Central Ground Water Board (CGWB), Ministry of Jal Shakti, Government of India [11]. The dataset is based on the years 1991 to 2020 and contains hydro-climatic variables, which include rainfall, temperature, relative humidity, evaporation, groundwater pumping index, recharge indicators, and geographical features, including latitude and longitude which play a significant role in the dynamics of groundwater.

The information in this research is arranged chronologically by each monitoring station to maintain the time features of groundwater time-series data. The data is preprocessed and then organized into station-based sequential samples that can be used in time-series forecasting models.

Time-based data splitting strategy is followed, with the previous observations being utilized in training and the most recent records being kept in testing. In particular, the model training is done using records before 2016 and the rest of the observations are used to test. This method prevents information leakage and is more realistic in terms of ground water forecasting in the real world.

All input features are z-score normalized to attain a zero mean and unit variance before the model is trained. This

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normalization makes all variables equally contribute to the optimization of the model and enhances the numerical stability of the training process.

B. Feature Transformation Using PCA and DWT

In order to eliminate the issue of multicollinearity and high dimensionality of the hydro-climatic variables, the Principal Component Analysis (PCA) is employed to normalize the feature matrix $\mathbf{X}$:

$$\mathbf{Z} = \mathbf{X}\mathbf{W} \tag{1}$$

where $\mathbf{W}$ represents the eigenvector matrix corresponding to the largest eigenvalues of the covariance matrix of $\mathbf{X}$. The principal components capturing 95% of the total variance are retained, which alleviates redundancy while preserving essential information.

The groundwater level time series is subjected to Discrete Wavelet Transform (DWT) to eliminate noise and isolate significant temporal patterns. The signal $x(t)$ is decomposed as:

$$x(t) = \sum_{k} a_{j,k}\phi_{j,k}(t) + \sum_{j}\sum_{k} d_{j,k}\psi_{j,k}(t) \tag{2}$$

where $\phi_{j,k}(t)$ and $\psi_{j,k}(t)$ are scaling and wavelet functions, respectively. The denoised signal reconstructed from the approximation coefficients serves as an additional input feature for sequence modeling.

The Daubechies wavelet (db4) is used to decompose the groundwater time series in this study because it is effective in capturing hydrological signals. The two level method is used to decompose the approximation and detail coefficients allowing the noise to be removed successfully and the important temporal trends to be retained. The denoised groundwater signal, reconstructed using the approximation coefficients is then taken as another feature in sequence modeling.

C. LSTM Network Architecture

The transformed features are clustered into fixed length sequences with a set lookback window of eight time steps, which allows the model to acquire temporal dynamics in groundwater dynamics. The sequences have the PCA-derived components alongside the denoised groundwater level signal obtained through DWT.

The forecasting model is developed based on a Long Short-Term Memory (LSTM) neural network with a single LSTM layer with 32 hidden units and fully connected regression layers. The network is trained with Adam optimizer with a learning rate of 0.002 and the loss is mean squared error (MSE).

Gradient clipping is employed to avoid exploding gradients, and early stopping, which is determined by validation loss, is employed to reduce overfitting and enhance generalization performance. The model is trained in mini-batches and the parameters that perform the best throughout the training process are saved to be evaluated finally.

The training is based on a lookback window of eight time steps to describe temporal relationships in groundwater dynamics. The data is split into training and validation sets in a 90-10 split and mini-batch gradient descent with a batch size of 256 is used in the optimization process. Early stopping is used to train the model up to 40 epochs to guarantee stable convergence and prevent overfitting.

D. Training and Validation Strategy

The model is trained using the Adam optimizer and the mean squared error (MSE) as a loss function and is as follows:

$$L = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{3}$$

where N is the number of samples; y_i is the measured groundwater level; and $\hat{y}_i$ is the model predicted value. This loss is more punitive to larger prediction errors, and thus it is appropriate to continuous-valued groundwater level prediction tasks.

Gradient clipping is used to stabilize training, and early stopping is used based on validation performance to avoid overfitting. The optimized model parameters and preprocessing components are stored together to enable reproducible inference and deployment.

V. Experimental Results and Performance Analysis

In this section, the experimental analysis of the proposed PCA-DWT-LSTM framework is introduced on the basis of quantitative and visual analysis. The aim is to determine model convergence, predictive accuracy and generalization under real world groundwater forecasting conditions.

A. Model Training Behavior

Fig. 2 indicates the convergence characteristics of the proposed model during training. The figure demonstrates the development of training and validation mean squared error (MSE) over epochs. The first epochs are characterized by a rapid decrease in the training loss, after which the training and validation curves converge. The close alignment among training and validation curves shows stable learning behavior suggesting that the model generalizes well to un-seen data without significant overfitting. This behavior highlights the effectiveness of the preprocessing pipeline and regularization strategies used in the proposed framework, which shows the appropriateness of the preprocessing and regularization strategies that are adopted.

B. Quantitative Performance Evaluation

The quantitative performance evaluation involves the evaluation of the performance based on measurable factors. The quantitative performance evaluation is the assessment of the performance in terms of measurable factors.

The quantitative performance of the proposed groundwater forecasting framework is assessed based on- standard regression measures such as mean absolute error (MAE), root mean

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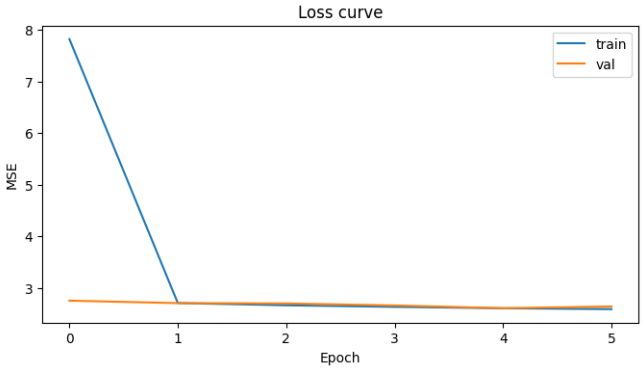


Fig. 2. Training and validation loss (MSE) convergence of the proposed PCA–DWT–LSTM model.

squared error (RMSE), and coefficient of determination (R^2). These measures, quantify the extent of prediction error and goodness of fit between predicted and observed ground water levels.

The convergence behaviour of the proposed model during the training process is presented in Fig. 2. The training and validation loss curves indicate a steady reduction in the mean squared error (MSE) with increase in the number of epochs indicating steady learning of temporal groundwater patterns. The fact that the training, and validation curves, are so close implies that the model is well-generalized and that it is not overfit.

A comparative analysis was also used to further assess the effectiveness of the suggested framework, and some of the baseline models used comprised standalone LSTM, PCA-LSTM, and DWT-LSTM architectures. The same dataset, lookback window, and optimization settings were used to train all models so that they could be fairly compared.

Table I provides the quantitative summary of the comparison. The single LSTM model has a decent forecasting ability but, when PCA is added, the model enhances its performance by eliminating redundancy and multicollinearity of hydro-climatic variables. Equally, the use of DWT improves the model in terms of multi-scale temporal patterns by removing noise in groundwater signals.

The suggested PCA-DWT-LSTM model has the lowest prediction errors and the largest coefficient of determination compared to all the reviewed models. These findings show that dimensionality reduction and wavelet-based signal processing with LSTM contribute greatly to the accuracy of groundwater level prediction.

TABLE I
COMPARATIVE PERFORMANCE OF BASELINE AND PROPOSED MODELS

Model	MAE	RMSE	R^2
LSTM	1.681	2.102	0.842
PCA-LSTM	1.529	1.973	0.871
DWT-LSTM	1.463	1.915	0.892
PCA-DWT-LSTM (Proposed)	**1.412**	**1.832**	**0.909**

In the case of the West Bengal regional subset, the model has a competitive performance with stable error levels, which show that the model can be generalized to spatially different hydro-climatic conditions. These findings indicate the strength of the hybrid PCA-DWT-LSTM model in predicting ground-water levels at various monitoring stations.

Residual analysis was done to further test the reliability of the predictions by the difference between the observed and predicted ground water levels. The residual distribution shows that the bulk of the prediction errors are concentrated around zero with low variance implying that the forecasting behaviour is consistent across varying groundwater regimes. This analysis gives an indirect confidence evaluation of the proposed model and shows that it is very strong in real world groundwater prediction situations.

C. Temporal Prediction Analysis

The forecasted groundwater levels are quite close to the temporal variations that are seen in the real measurements, and they are able to capture the seasonal patterns as well as the long-term variations, which can be observed in Fig. 3. Minor deviations are found when abrupt changes happen, and these changes can be caused by sudden hydrological events like recharge events or pumping variations. In general, the fact that the predicted and observed series are very close indicates the potential of the hybrid model to learn complicated temporal relations in groundwater dynamics.

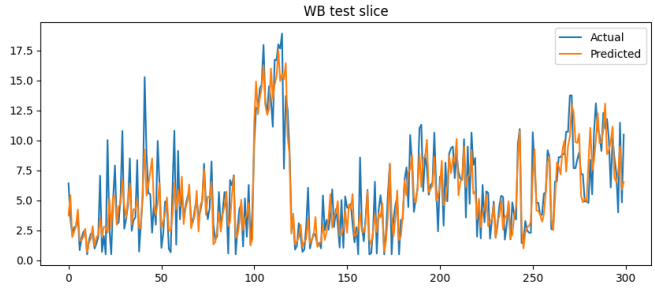


Fig. 3. Comparison of actual and predicted groundwater levels for the West Bengal test dataset.

D. Prediction Accuracy Distribution

The scatter distribution presented in the Fig. 4 demonstrates that there is high correlation between the predicted and observed groundwater levels with the majority of the points clustering around the diagonal reference line. This trend proves the high predictive power of the suggested model and justifies the values of R^2 in Table I. Also, the lack of systematic dispersion indicates that the model is consistent both in the low and high groundwater level ranges.

E. Discussion

The joint application of PCA and DWT with LSTM to predict groundwater levels has been reported to have some of the best accuracies ever. PCA enhances the representation of the hydroclimatic variables and minimizes multicollinearity, whereas DWT enhances the quality of the signal by removing

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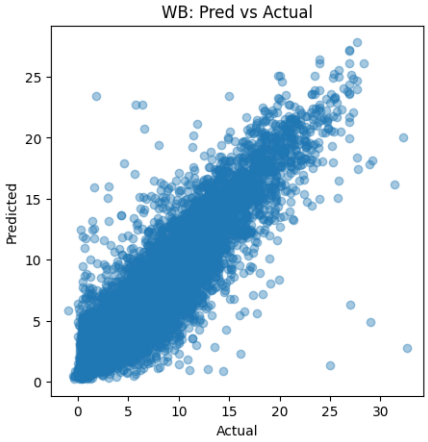


Fig. 4. Predicted against actual scatter plot and actual ground water level of the West Bengal test dataset.

noise and capturing multi-scale varying signals over time. Improvements in the forecasting of groundwater levels with the help of wavelet LSTM frameworks have also been reported in previous works [4], [6], [13] and the application of PCA based dimensionality reduction enhanced the accuracy of the multivariate groundwater prediction [3], [7]. These findings support the usefulness of hybrid machine learning methods to model complex hydroclimatic relationships, and this is in line with earlier research on time-series forecasting with machine learning models [16].

VI. Conclusion and Future Scope

To address the nonlinear, noisy and temporal characteristics of the groundwater systems, the paper presented a hybrid groundwater level forecasting model that integrates Principal Component Analysis (PCA) with Discrete Wavelet Transform (DWT) and Long Short-Term Memory (LSTM) networks. The proposed approach is a combination of dimensionality reduction, signal denoising, and deep temporal learning, which works well to ensure that long-term patterns and seasonal changes that are present in groundwater time-series data are represented. Experimental analysis shows that the proposed PCA-DWT-LSTM framework is more accurate and strong in its prediction capability compared to the conventional standalone forecasting models. PCA reduces redundancy and multicollinearity of hydro-climatic variables and DWT maximizes quality of signals in groundwater time-series data through reduction of noise. These altered properties enable the LSTM network to learn more important temporal relationships, and thus the groundwater level is more accurately predicted. The results indicate that the proposed framework can be used to support groundwater monitoring and decision-support systems in sustainable water resource management. It is possible to conduct future research to include real-time sensor data and climate projection variables to increase the accuracy of long-term forecasting. In addition, the uncertainty can be quantified and explainable methods of artificial intelligence can be presented in order to make the models more transparent and assure the stakeholders. The framework can also be extended to regional ground water assessment and embedded in web-based decision-support systems to aid ground water management by the policymakers and water resource managers.

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